

Keys Without Trust Assumptions: Integrating Quantum Key Distribution and Post-Quantum Cryptography into the Internet Protocol Stack

A CN1 analysis of the QKD control/data plane, ETSI key-delivery API, and hybrid IKEv2 key establishment

Truong Phat Tu • **Computer Networking and Cyber Security I — CSC-6440-1** • **Fall 2026** • **Visvasuresh Govindaswamy**

Emerging Networking Technology / Network Security | #WGP — From Learning to Building — From Building to Innovation

01 PROBLEM & BACKGROUND

THE NETWORKING PROBLEM

Today's TLS, IPsec/IKEv2, SSH, and VPN sessions rely on key exchange based on factoring or discrete-log assumptions. A sufficiently capable fault-tolerant quantum computer could undermine these foundations.

HARVEST NOW / DECRYPT LATER: captured ciphertext can threaten long-lived medical, financial, infrastructure, and archival data.

WHY THIS IS A NETWORK PROBLEM

Changing key agreement alters handshakes, message sizes, round trips, MTU/fragmentation, latency, state machines, and middlebox compatibility.

THE TWO CANDIDATES

PQC: software-scale reach over ordinary IP; relies on new computational assumptions.

QKD: derives keys from quantum-channel measurement statistics; requires specialized hardware, links, and operational trust.

Neither is a universal drop-in answer.

MINIMUM CONCEPTS

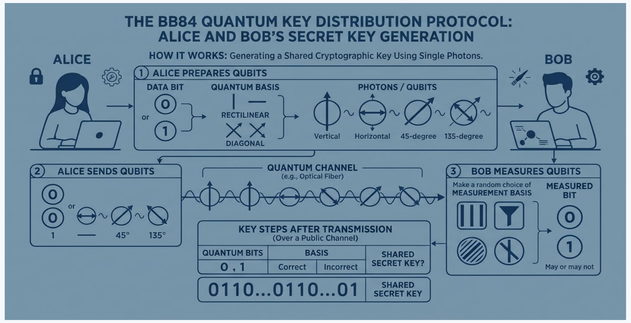
- **AES-256 bulk encryption remains symmetric; key agreement is the central quantum-risk focus.**
- **Measurement disturbance + no-cloning make interception statistically detectable.**
- **QBER estimates error; above the protocol threshold, abort key generation.**
- **QKD needs a quantum channel AND a public, authenticated classical channel.**

MOTIVATING QUESTION

How does a quantum-derived key reach an application, IPsec tunnel, or router—and what does it cost in latency, throughput, topology, and trust?

KEY DISTINCTION

QKD produces key material; it is not an encryption algorithm. It does not authenticate by itself and does not replace the Internet. It feeds keys into existing protocols.



POST-QUANTUM CRYPTOGRAPHY

PQC runs on ordinary CPUs and IP networks.

- **ML-KEM — key encapsulation (FIPS 203)**
- **ML-DSA — digital signatures (FIPS 204)**
- **SLH-DSA — hash-based signatures (FIPS 205)**

POST-QKD / HYBRID (this poster): QKD supplies link-specific key material; PQC supplies scalable authentication and an independent key-agreement path. Combine independent contributions for defense in depth.

02 CN1 CONCEPTS & WORKFLOW

CN1 CONNECTIONS

PROTOCOL / STATE MACHINES — transmission → sifting → reconciliation → estimation → privacy amplification → confirmation.

ENCAPSULATION — the opaque key_ID moves in HTTP/JSON over TLS/TCP/IP; the secret is delivered locally by each KME.

SECURITY ARCHITECTURE — trust shifts from mathematical assumptions toward devices, facilities, and relay nodes.

THROUGHPUT • DELAY • QUEUING

Secret-key rate falls with fiber loss/distance. QKD keys seed or rekey AES rather than encrypting every bit.

Post-processing adds delay; advance key buffering converts latency into a storage/queueing problem.

KME key pools are producer/consumer queues. Demand above supply can cause key starvation.

LOSS • ADMISSION • TOPOLOGY

Photon loss is expected; sifting retains detected, basis-matched events. Reconciliation corrects key mismatch.

Multiple SAEs require allocation/admission control; oversubscription can return HTTP 503. No conventional store-and-forward routing of qubits. Reach uses trusted-node chains, optical switching, or satellite links.

QKD WORKFLOW

0 Bootstrap authentication
Pre-shared secret or PQC signatures; authenticate classical channel.

1 Quantum transmission
Alice encodes random bits/bases; Bob measures. Most pulses may be lost.

2 Sifting
Reveal bases—not bit values; discard mismatches.

3 Parameter estimation
Sample bits to estimate QBER; abort if unsafe.

4 Reconciliation
Correct residual mismatches; account for public leakage.

5 Privacy amplification
Hash/compress to a shorter final secret.

6 Confirm + store
Verify agreement; place key in KME pool indexed by key_ID.

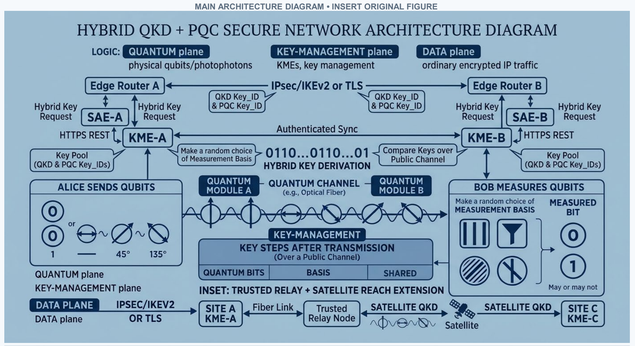
7 Deliver + consume
ETSI REST obtains keys/IDs; hybrid handshake derives session key.

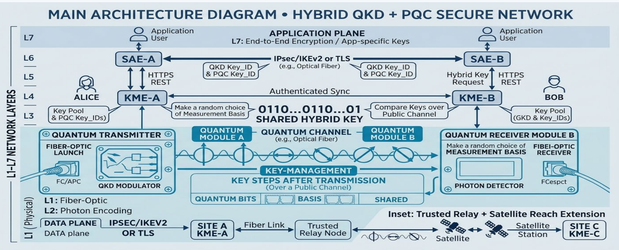
KEY INSIGHT: QKD is generated at L1, orchestrated at L7, and consumed at L2/L3/L4.

03 ARCHITECTURE & LAYER MAPPING

HYBRID QKD + PQC SECURE NETWORK

MAIN ARCHITECTURE DIAGRAM • INSERT ORIGINAL FIGURE





PROTOCOL-LAYER MAPPING

L7 / Application	ETSI GS QKD 014 REST/JSON; SAE ↔ KME; status, key request, key redemption.
L5–L6 / Session	TLS handshake and key schedule; session rekey based on pool supply.
L4 / Transport	TCP carries REST calls and interactive post-processing.
L3 / Network	IPsec/IKEv2: consume hybrid key material; trust-aware path selection.
L2 / Link	Optical/Ethernet encryptors may draw AES keys from KME.
L1 / Physical	Quantum pulses on dark fiber, dedicated wavelength, or free-space optics.
Conceptual plane	Quantum network may be modeled as a distinct layered plane.

HYBRID SESSION KEY (CONCEPT)

KDF(classical ECDHE || ML-KEM shared secret || QKD key)

IKEv2 extensions: RFC 8784 / RFC 9370

Security goal: final key depends on independent contributors; operational details must follow protocol specifications.

04 EVIDENCE, TRADE-OFFS & OUTLOOK

REAL-WORLD EVIDENCE

- **SwissQuantum, Geneva (~20 km; 2008)**
- **Tokyo QKD Network (2010; multi-vendor testbed)**
- **SECOQC, Vienna (2008; trusted-node network)**
- **UK research network (2020; trusted-node-free demonstration)**
- **CSIRO testbed (2024)**
- **Micius / space-to-ground demonstrations**

Published ETSI interfaces and IKEv2 integration work connect testbeds to networking practice.

BENEFITS

- **Information-theoretic key secrecy (within stated assumptions)**
- **QBER provides measurable channel telemetry**
- **Hybrid defense in depth**
- **API abstraction can aid modular integration**
- **Fits fixed, high-value links**
- **Fresh key supply supports frequent rekeying**

LIMITATIONS

- **Distance / key-rate constraints**
- **Trusted relays expose key in plaintext**
- **Authentication bootstrap required**
- **Dedicated fiber, detectors, cost, operations**
- **No IP-like routing or multicast**
- **Side channels / device imperfections**
- **Pool exhaustion & interoperability**
- **Energy overhead**

CRITICAL PERSPECTIVE

NSA does not support QKD for National Security Systems; UK NCSC generally prioritizes PQC for most settings. Include agency guidance and verify its current wording before submission.

STATUS SNAPSHOT

QKD physics: mature / established
Commercial metro QKD: niche deployment
ETSI / ITU frameworks: published standards
PQC (ML-KEM, ML-DSA, SLH-DSA): standardized
Hybrid QKD+PQC IKEv2: emerging / early integration
Quantum repeaters: research

FUTURE IMPACT

1–3 years: PQC becomes default; QKD premium option on selected links.
3–7 years: key-as-a-service, pool telemetry, crypto-agility.
7–15 years: wider reach depends on practical quantum repeaters; satellites may bridge distance.

CONCLUSION

The decisive challenge is delivering and consuming keys under real constraints—not merely generating them. QKD relocates trust; it does not remove it. Likely near-term pattern: PQC for reach/authentication, QKD where justified, and hybrid key schedules.

APPLICATION SCENARIO

University medical research center ↔ off-site data center (~15 km dark fiber)
Long-lived genomic data • SAE VPN gateways • ETSI key delivery • hybrid IKEv2 rekey • PQC-only fallback if pool unavailable

<<FILL: deployment diagram, local assumptions, or project-specific result>>

PROJECT FINDING / TAKEAWAY:

Conclusion this research showing the necessary of quantum key distribution and post quantum key distribution as hybrid security layers apply in Protocol data transfer in network system in protect data out of of concept **Harvest Later** and using **No – Cloning** theorem.

SELECTED REFERENCES | Complete, verify, and renumber before submission

[1] Bennett & Brassard, IEEE Conf. (1984). [2] Ekert, Phys. Rev. Lett. (1991). [4] ETSI GS QKD 014 V1.1.1 (2019). [6] IEEE JSAC hybrid integration (2025; complete author/pages). [7] NIST FIPS 203 (2024). [8] NIST FIPS 204 (2024). [9] IETF RFC 8784 (2020). [10] IETF RFC 9370 (2023). [11] ITU-T Y.3800 (2019). [12] Liao et al., Nature (2017). [14] Lucamarini et al., Nature (2018). [18] NSA guidance (verify current citation). [19] UK NCSC (2020).